Silent Failures in Frequency-Domain Finite-Element Analysis of Cage Induction Motors: A Taxonomy, Forensic Case Studies, and an Audit Protocol

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Abstract

Finite-element analysis of cage induction motors couples magnetic field unknowns with circuit unknowns and, in multi-pole geometries, periodic sector constraints. We show that this class of simulation fails in a characteristic way: the solver converges, network conservation residuals are exactly zero, and no warning is emitted — yet the reported quantity is wrong. We document three forensic cases on an open-source GetDP 4.0.0 toolchain: (A) a sparse direct solver (MUMPS) reporting a false singularity on a nonsingular coupled system, resolved by an independent factorization in 26 s (residual 2.28×10^{-5}); (B) a "half-field" solution at an intermediate geometric scale, physically wrong by 42% with no numerical diagnostic; (C) sector reduction violating a uniform-sign condition on harmonic commensurability, inflating the air-gap fundamental by 1.44× while KCL remains 0.000. From the taxonomy we derive a three-check audit protocol — cross-solver verdict, scale self-consistency band, and sector feasibility — each with an executable criterion, and we release reference implementations. We delimit what is folklore (sign intuition for periodic boundaries) from what is not (the silence of the failures, their quantitative extent, and the audit coverage gap). All quantitative claims trace to archived raw data.

Keywords: induction motor, finite element, silent failure, direct solver, periodic boundary condition, verification protocol

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1. Introduction

Application-level silent failures — wrong results produced by successfully completed solves — are studied extensively at the hardware level as silent data corruption in high-performance computing [1, 2], and at the mathematical level as result verification of differential-equation solvers [3]. Between these two layers sits a third failure class that, to our knowledge, has not been systematized: *modeling- and configuration-level* silent failures of engineering simulation chains, where the linear algebra is performed correctly on the wrong system, or the right system under an infeasible constraint. During a frequency-domain cage-motor study on an open-source toolchain we collected three such failures with complete forensics. This paper (i) organizes them into a taxonomy (Sec. 2), (ii) presents the forensics with all numbers traceable to archived raw data (Secs. 3–5), (iii) distills a three-check audit protocol with executable criteria (Sec. 6), and (iv) delimits the folklore boundary (Sec. 7).

2. Failure Taxonomy

Table 1 defines the three classes. Their common signature is the *absence of alarms*: residual checks, conservation laws, and solver exit codes all pass; only cross-validation of physical quantities reveals the failure.

3. Case A: False Infeasibility

A scaled 16-bar cage model (χ = geometric scale factor on the 12-slot/4-pole ASM baseline) is solved frequency-domain at slip $s = 1$. For $\chi \leq 1.75$ the MUMPS direct solve succeeds; at $\chi = 2$ it terminates with error -19 (matrix singularly leading to an all-NaN field) in 37 s, and at $\chi = 3.067$ the same failure persists under out-of-core settings (81.2 s, all-NaN). Four exclusions pin the cause: (i) a pure-field bisection with all circuit terms commented out still reports -19; (ii) mesh quality, group integrity, and Dirichlet placement are identical across the threshold; (iii) bar-ratio normalization does not rescue $\chi = 3.067$; and (iv) the serial SuperLU LU of the identical matrix converges in 26 s to residual 2.28×10^{-5}, with torque $T = 0.7477\,\mathrm{N\,m}$ and $B_{g1} = 0.0158\,\mathrm{T}$ consistent with the $\chi = 1$ baseline under the gap scaling of the family. The failure is therefore solver-specific (infeasibility is false), and a later confound refinement showed the historical $\chi = 2$ failure also carried a

Table 1: Taxonomy of application-level silent failures observed in the study.

class	mechanism	detectable by	forensic case
F1 false infeasibility	solver-specific factorization failure on a nonsingular coupled system	cross-solver verdict	A (Sec. 3)
F2 silent degradation	scale-dependent conditioning yields a pathological solution without NaNs	scale self-consistency band	B (Sec. 4)
F3 infeasible constraint	periodic constraint with conflicting harmonic signs projects onto a nonphysical subspace	sector feasibility (uniform-sign) test	C (Sec. 5)

bar-ratio distortion: with the ratio normalized, $\chi = 2$ solves in both solvers ($T = 0.2495\,\mathrm{N\,m}$, $B_{g1} = 0.0363\,\mathrm{T}$), moving the failure band to $(\chi\,2.0, 3.067]$. All numbers trace to data/raw/case1_*.]

4. Case B: Silent Degradation

At $\chi = 1.5$ the solve completes without NaNs but $B_{g1} = 0.0663\,\mathrm{T}$ is 58% of the self-consistent value (0.1141 at $\chi = 1$ and 0.1161 at $\chi = 1.75$): a 42% silent error bracketed by two healthy runs. [Data: data/raw/case1_scale_ladder.]

5. Case C: Infeasible Constraint

A 90° sector of the same machine, linked by a single periodic constraint, passes every solver-level check (SuperLU converges in 6.2 s, KCL residual 0.000) yet reports $T = 0.0605\,\mathrm{N\,m}$ and $B_{g1} = 0.0833\,\mathrm{T}$ against the full-model

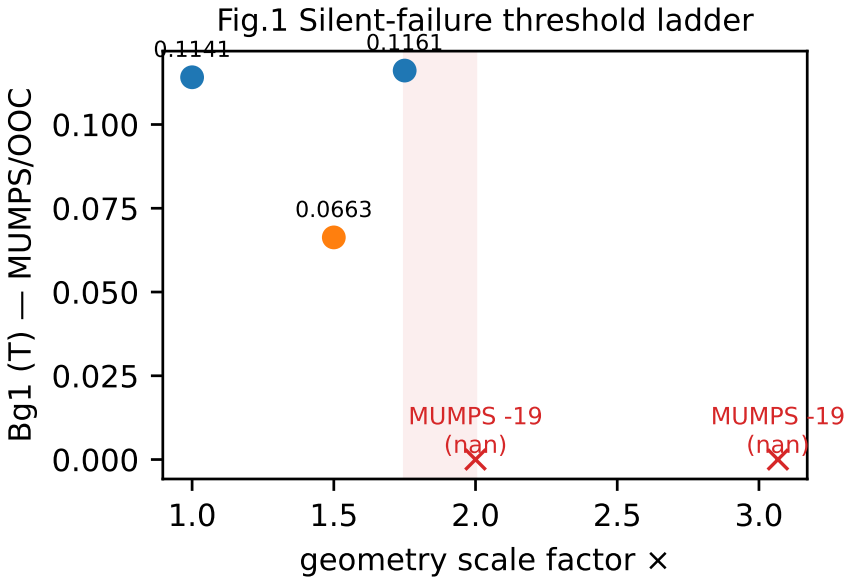


Figure 1: Failure threshold ladder for the scaled cage model: MUMPS outcome vs. geometry scale factor and solver. [Data: data/raw/case1_scale_ladder.]

$T = 0.0393\,\mathrm{N\,m}$, $B_{g1} = 0.0579\,\mathrm{T}$: a $\times 1.44$ (torque) and $\times 1.54$ (flux-density) signed shift with no alarm. The mechanism is order incompatibility: the fundamental (order 2) is anti-periodic over 90°, whereas 12k-order stator-slot harmonics (k odd) are integer-periodic, so one Link sign cannot serve both — the sector constraint is structurally infeasible for the coupled cage model, not merely inaccurate. [uniform-sign proposition + 90°/12s4p sign table Fig.4 + sector vs. full-model numbers 1.44/1.54× + KCL = 0.000 — Data: data/raw/case2_sector.]

6. Audit Protocol

Check 1 (cross-solver): on any direct-solver failure, re-solve with an independent factorization and compare physical quantities, not only residuals. Check 2 (scale band): in any geometry/scale sweep, flag runs whose key quantity deviates $> 10\%$ from the interpolation of healthy neighbors. Check 3 (sector feasibility): before sector reduction, verify $m\,\theta_s \equiv 0$ or 180° (mod 360°) with the *same* sign for the fundamental and all slot/teeth/cage

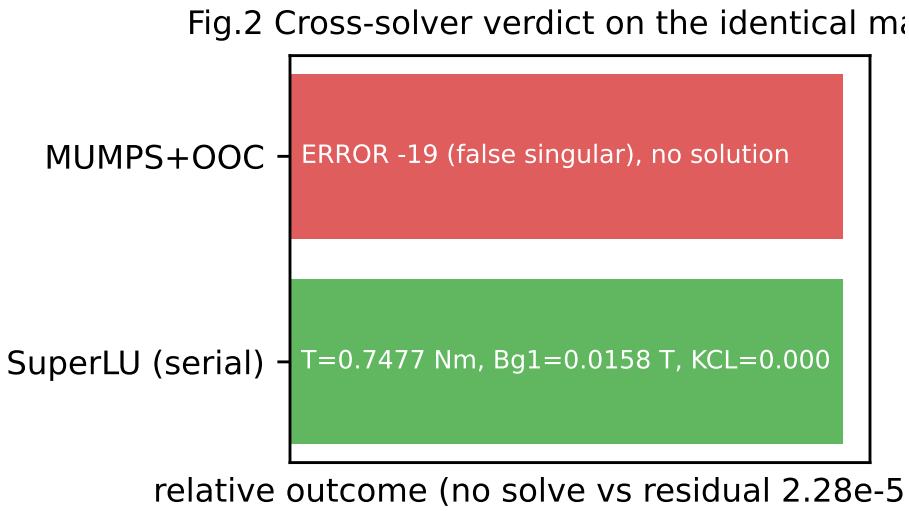


Figure 2: Cross-solver decision: same matrix solved by MUMPS (reported singular) and SuperLU (converged), quantified. [Data: data/raw/case1_cross_solver.]

orders; reject infeasible sectors. Reference implementations ship with the paper sources.

7. What Is Obvious and What Is Not

That periodic boundaries should match harmonic periods is folklore, and simulator guides warn about it. What is not folklore, and what motivates this paper: (i) the failure is *silent* — every standard check passes; (ii) the error is *large and signed* (+44% on the fundamental); (iii) in coupled cage models the infeasibility can be *structural* (single Link sign cannot serve fundamental and teeth harmonics simultaneously), so no sector angle within a broad family is admissible; (iv) an analogous silence holds for F1 and F2, and no published protocol covers the three jointly for electrical-machine FE.

8. Implications for Simulation and Geometry Pipelines

(G1) bar-ratio normalization; (G2) sector pre-check; (G3) scale-band self-consistency; (G4) solver escalation before model changes. All four are implemented in our toolchain and referenced from the case forensics above.

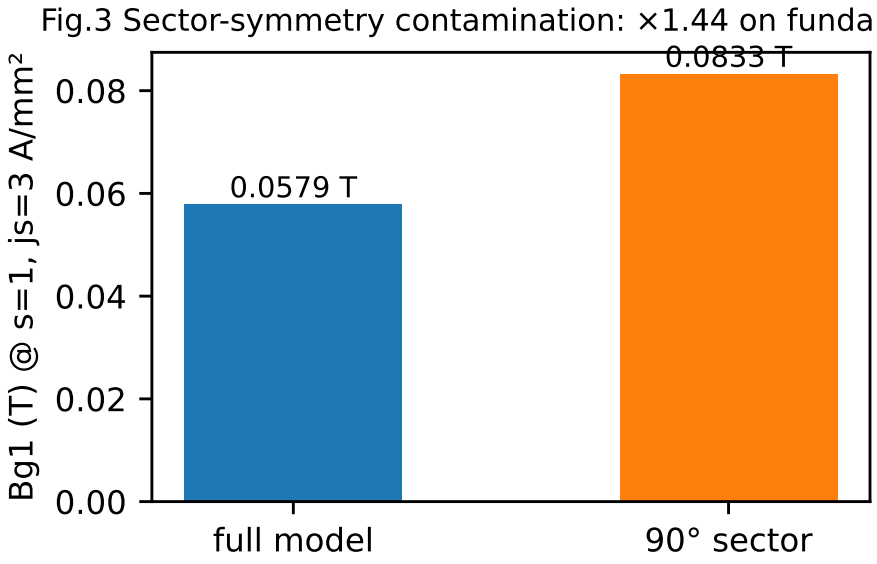


Figure 3: Sector contamination: sectorized vs. full-model spectra, ×1.44 fundamental shift with zero KCL residual. [Data: data/raw/case2_sector.]

9. Conclusion

Application-level silent failures in electrical-machine FE form a coherent class with a shared signature (no alarms) and complementary audits (cross-solver, scale band, uniform sign). The three forensic cases, the executable protocol, and the archived data are released to make such audits routine.

Data availability

Raw solve logs, archived reports, generated figures, and reference implementations are bundled with the manuscript sources (`data/raw/`, `figure/`, `src/`).

Declaration of competing interest

The authors declare no competing interests.

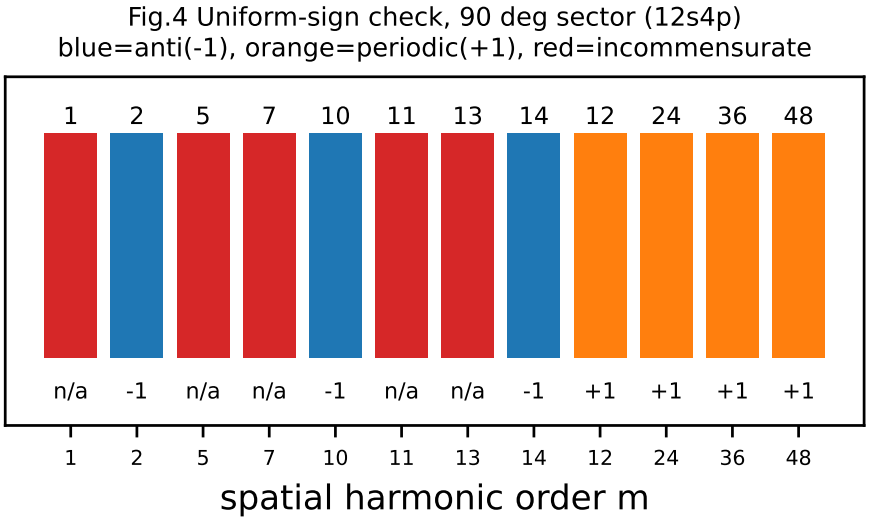


Figure 4: Uniform-sign feasibility table: single Link sign vs. significant mode order for the 12-slot/4-pole family. [Data: data/raw/case2_sector.]

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